

Symmetric Pattern

Custom Feature for Onshape - linear and grid patterns with symmetric margins.

What It Does

Symmetric Pattern creates copies of a selected body or face operation between boundary vertices. Linear mode distributes columns between Start point and End point. Grid mode adds rows using Row end point as the row boundary, keeping first and last row margins equal. Optional stagger reduction shortens shifted rows from the right side.

Inputs

Input	Purpose
Pattern type	Choose Body for whole parts or Faces for selected operation faces such as holes.
Start point	Boundary vertex for the first column and first row reference.
End point	Boundary vertex for the column direction.
Columns	Total number of columns, including the original feature.
Mode	Linear for one row, Grid for multiple rows.
Rows	Total number of rows in Grid mode.
Row end point	Boundary vertex for the row direction. Use a corner or vertex on the intended grid boundary.
Stagger	Optional horizontal offset for every second row: None, 50%, or Custom percentage.
Reduce staggered rows	When Stagger is active, shorten shifted rows by the Reduce by value. Elements are removed from the right end on

Basic Workflow

- 1 Create the original feature, for example one cut or hole.
- 2 Add Symmetric Pattern from the custom feature toolbar.
- 3 Select the target body or target faces.
- 4 Select Start point and End point to define the column boundary.
- 5 Set Columns. Use Linear mode for one row.
- 6 For Grid mode, set Rows and select Row end point on the opposite row boundary.
- 7 Use Stagger when alternating rows should be shifted horizontally.
- 8 If shifted rows enter the right margin, enable Reduce staggered rows and set Reduce by.

Grid Placement Rule

Columns and rows use the same margin rule. The tool measures the original feature position, computes the margin to the selected boundary, and mirrors that margin at the opposite boundary. Row end point is a boundary point, not the center of the last copy. Stagger reduction removes elements only from the right end of shifted rows, so the first shifted element keeps its stagger offset.

Recommended Test Before Publishing

Test Linear mode with 4-5 columns. Then test Grid mode with 2, 3, 4, 5, and 7 rows. For Stagger, test Columns 5, Reduce staggered rows on, Reduce by 1: shifted rows should have 4 elements and keep their left stagger offset.

Version 1 - prepared for Onshape custom feature publication.